Commutative Algebra

Shing Tak Lam

Michaelmas 2023*

Contents

1 Chain conditions **1**

2 Tensor products **4**

2.1 Tensor product of R-linear maps 9

Lecture 1

In this course, a ring is a commutative unital ring R. One non-commutative exception is the ring $\mathrm{End}(M)$, where M is an abelian group. This is a ring with pointwise addition, and composition as multiplication.

Definition 0.1 (module)

An *R-module* M is an abelian group M with an fixed ring homomorphism $\rho : R \to \mathrm{End}(M)$. We will write $r \cdot m := \rho(r)(m)$.

Remark 0.2. By definition, this implies that $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$, $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$ and $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$.

Example 0.3 (Examples of modules)

- Let k be a field. Then a k-module is the same as a k-vector space.
- Every abelian group is a $\mathbb{Z}$-module in a unique way, since we must have that $\rho(1) = \mathrm{id}_M$. Therefore, abelian groups and $\mathbb{Z}$-modules are the same thing.
- Every ring R is (trivially) an R-module.
- More generally, $R^{\oplus \mathbb{N}}$ (direct sum) and $R^{\mathbb{N}}$ (direct product) are R-modules.

1 Chain conditions

Definition 1.1 (Noetherian, Artinian module)

An R-module M is *Noetherian* if one of the following (equivalent) conditions hold:

1. Every ascending chain of submodules $M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots$ stabilises. That is, it is eventually constant.
2. Every non-empty set Σ of submodules of M has a maximal element.

M is *Artinian* if we replace in the above: ascending with descending, maximal with minimal.

*Based on lectures by Oren Becker. Last updated October 10, 2023.

Lemma 1.2. An R-module M is Noetherian if and only if every submodule of M is finitely generated.

In particular, every Noetherian module is finitely generated. If $R = \mathbb{Z}[T_1, T_2, \dots]$, with $M = R$ as an R-module. Then M is finitely generated. On the other hand, $M' = \langle T_1, T_2, T_3, \dots \rangle$, is not finitely generated.

Definition 1.3 (Noetherian, Artinian ring)
A ring R is Noetherian (resp. Artinian) if it is Noetherian (resp. Artinian) as an R-module.

Example 1.4
1. $\mathbb{Z}$ is Noetherian (as it is a PID), but not Artinian (e.g. $\langle 2 \rangle \supseteq \langle 4 \rangle \supseteq \langle 8 \rangle \supseteq \cdots$).
2. $\mathbb{Z}[1/2]/\mathbb{Z}$ is Artinian, but not Noetherian as a $\mathbb{Z}$-module.
3. A ring R is Artinian if and only if R is Noetherian and R has Krull dimension 0.

Definition 1.5 (Exact sequence)
A sequence

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots$$

of R-modules and R-module homomorphisms is *exact* if $\operatorname{im}(f_i) = \ker(f_{i+1})$ for all i.

Definition 1.6 (Short exact sequence)
A *short exact sequence* (SES) is an exact sequence of the form

$$0 \longrightarrow N \xhookrightarrow{\iota} M \twoheadrightarrow L \longrightarrow 0$$

That is, we have an embedding $\iota : N \hookrightarrow M$, and an isomorphism $L \cong M/\iota(N)$.

Lemma 1.7. Let

$$0 \longrightarrow N \longrightarrow M \longrightarrow L \longrightarrow 0$$

be an SES of R-modules. Then M is Noetherian (resp. Artinian) if and only if N and L are Noetherian (resp. Artinian).

Proof. We may assume without loss of generality that N is a submodule of M. Let $P_1 \subseteq P_2 \subseteq \dots$ be an increasing (resp. decreasing) sequence of submodules of M. In this case,

$$N \cap P_1 \subseteq N \cap P_2 \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of N, hence eventually constant. Similarly,

$$\frac{N + P_1}{N} \subseteq \frac{N + P_2}{N} \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of $L = M/N$, hence eventually constant. For large n, we will have

$$P_n \subseteq P_{n+1} \quad N \cap P_n = N \cap P_{n+1} \quad N + P_n = N + P_{n+1}$$

Hence $P_n = P_{n+1}$ for large enough n. □

Corollary 1.8. If $M_1, \dots, M_n$ are Noetherian (resp. Artinian) R-modules, then $M_1 \oplus \cdots \oplus M_n$ is Noetherian (resp. Artinian).

Proof. By the lemma and induction. □

Recall a module homomorphism

$$\varphi : M_1 \oplus \cdots \oplus M_n \to L$$

is the same as a collection of module homomorphism $\varphi_i : M_i \to L$. This is also true for infinite direct sums (but not products!).

Proposition 1.9. For a Noetherian (resp. Artinian) ring R, every finitely generated R-module is Noetherian (resp. Artinian).

Proof. M is finitely generated if and only if there exists a surjection $R^n \twoheadrightarrow M$ for some $n \in \mathbb{N}$. The fact that R^n is Noetherian (resp. Artinian) implies that M is Noetherian (resp. Artinian), as quotients of Noetherian (resp. Artinian) modules are Noetherian (resp. Artinian). □

Definition 1.10 (algebra)

An *R-algebra* A is a ring A with a fixed ring homomorphism $\rho : R \to A$. We will write $r \cdot a := \rho(r)a$.

Remark 1.11. Every R-algebra is an R-module.

Example 1.12

The polynomial ring $k[T_1, \dots, T_n]$ is a k-algebra. Do note however that it is a finitely generated by $T_1, \dots, T_n$ as a k-algebra, but it is infinite dimensional as a k-vector space.

Definition 1.13 (algebra homomorphism)

$\varphi : A \to B$ is an *R-algebra homomorphism* if φ is a ring homomorphism and $\varphi(r \cdot 1_A) = r \cdot 1_B$.

Definition 1.14 (finitely generated algebra)

An R-algebra A is *finitely generated* if there exists a surjective R-algebra homomorphism $R[T_1, \dots, T_n] \twoheadrightarrow A$ for some $n \in \mathbb{N}$.

Lecture 2

Theorem 1.15 (Hilbert basis theorem)**.** Every finitely generated algebra A over a Noetherian ring R is Noetherian (as a ring).

For example, if k is a field, then $k[T_1, \dots, T_n]$ is Noetherian.

Proof. It suffices to prove for $A = R[T_1, \dots, T_n]$, since every finitely generated algebra is a quotient of $R[T_1, \dots, T_n]$. Moreover, by induction, suffices to prove the result for $A = R[T]$.

Let $\mathfrak{a}$ be an ideal of $A = R[T]$. For every $i \geq 0$, define

$$\mathfrak{a}(i) = \left\{ c_0 \mid c_0 t^i + \cdots + c_i t^0 \in \mathfrak{a} \right\}$$

for the set of all leading coeffients of elements of degree i in $\mathfrak{a}$ (and containing 0). In this case, $\mathfrak{a}(i) \subseteq R$ is an ideal, and we have an ascending chain of ideals

$$\mathfrak{a}(i) \subseteq \mathfrak{a}(i+1) \subseteq \cdots$$

Since R is Noetherian, each $\mathfrak{a}$ is finitely generated (as an ideal), and the ascending sequence of ideal stabilises. That is,

$$\mathfrak{a}(m') = \mathfrak{a}(m)$$

for all $m' \geq m$. We write $\mathfrak{a}(i) = \langle b_{i,1}, \ldots, b_{i,m_i} \rangle$, where $b_{i,j} \in R$. Let $f_{i,j} \in \mathfrak{a}$ be a polynomial of degree i, with leading coefficient $b_{i,j}$. Define the new ideal

$$\mathfrak{b} = \left\langle f_{i,j} \mid i \leq m, 1 \leq j \leq m_i \right\rangle \trianglelefteq R[T]$$

In this case, $\mathfrak{b}(i) = \mathfrak{a}(i)$ for all i. By construction, $\mathfrak{b} \subseteq \mathfrak{a}$.

Suppose for contradiction that $\mathfrak{a} \nsubseteq \mathfrak{b}$. Take $f \in \mathfrak{a} \setminus \mathfrak{b}$ of minimal degree i. But $\mathfrak{b}(i) = \mathfrak{a}(i)$, and so there exists $g \in \mathfrak{b}$, of degree i, and with the same leading coefficient as f. That is, $\deg(f-g) < i$. By minimality, $f - g \in \mathfrak{b}$, and so $f = (f-g) + g \in \mathfrak{b}$. Contradiction. □

Therefore, if we have a subset $S \subseteq R[T_1, \ldots, T_n]/I$, then $\langle S \rangle = \langle S_0 \rangle$, where $S_0 \subseteq S$ is finite.

2 Tensor products

Let M, N be R-modules. An informal definition of their tensor product is

$$M \otimes_R N = \left\{ \sum_{i=1}^{\ell} m_i \otimes n_i \;\middle|\; m_i \in M, n_i \in N \right\}$$

where we have the relations $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$, $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$, and that for $r \in R$, $(rm) \otimes n = r(m \otimes n) = m \otimes (rn)$.

For example, consider $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3$. Then

$$x \otimes y = (3x) \otimes y = x \otimes (3y) = x \otimes 0 = 0$$

and so, $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3 = 0$. On the other hand, if we have vector spaces, then

$$\mathbb{R}^m \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong \mathbb{R}^{m\ell}$$

Recall $f : M \times N \to L$ is *R-bilinear* if $n \mapsto f(m_0, n)$ and $m \mapsto f(m, n_0)$ are R-linear for all $m_0 \in M, n_0 \in N$.

Definition 2.1 (tensor product of modules)

Let M, N be R-modules, let

$$\mathcal{F} = R^{\oplus(M \times N)} = \operatorname{span}_R \left\{ e_{(m,n)} \mid m \in m, n \in N \right\}$$

be the free module indexed by $m \times n$, and define $\mathcal{K} \subseteq \mathcal{F}$ for the submodule generated by the relations (where we write (m, n) for $e_{(m,n)}$)

$$\begin{aligned} (m, n_1) + (m, n_2) &= (m, n_1 + n_2) \\ (m_1, n) + (m_2, n) &= (m_1 + m_2, n) \\ r(m, n) &= (rm, n) \\ r(m, n) &= (m, rn) \end{aligned}$$

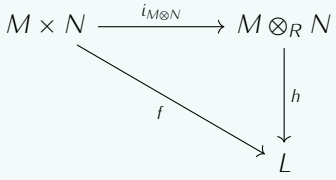
The *tensor product* is

$$M \otimes_R N := \frac{\mathcal{F}}{\mathcal{K}}$$

We have an R-bilinear map

$$\begin{aligned} i_{M\otimes N} : M \times N &\to M \otimes_R N \\ (m, n) &\mapsto m \otimes n \end{aligned}$$

Proposition 2.2 (universal property of tensor product)**.** For every R-module L and any R-bilinear map $f : M \times N \to L$, there exists a unique R-linear $h : M \otimes N \to L$, making the diagram



commute.

Proof. Uniqueness is clear, since we must have that

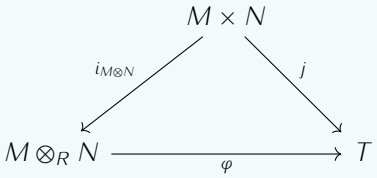
$$h(m \otimes n) = f(m, n)$$

since the pure tensors generate, h must be unique, if it exists. Therefore, suffices to show the above extends to an R-linear map $M \otimes_R N \to L$. This follows from the map

$$\begin{aligned} R^{\oplus(M\times N)} &\to L \\ e_{(m,n)} &\mapsto f(m, n) \end{aligned}$$

extending to a linear map (by the universal property of the direct sum), and that this map vanishes on $\mathcal{K}$. Therefore, h extends to $M \otimes_R N$ from the pure tensors. □

Proposition 2.3. Let M, N be R-modules, T an R-module, $j : M \times N \to T$ an R-bilinear map, (T, j) satisfying the universal property of tensors. Then there exists a unique R-linear isomorphism $\varphi : M \otimes N \to T$, such that



commutes.

Proof. By the universal property of tensor product, such a map φ exists, with $\varphi(m \otimes n) = j(m, n)$. Similarly, we have a homomorphism $\psi : T \to M \otimes_R N$. In particular,

$$\psi \circ \varphi \circ i_{M\otimes N} = i_{M\otimes N} = \mathrm{id}_{M\otimes N} \circ i_{M\otimes N}$$

In particular, by uniqueness in the universal property, we must have that $\psi \circ \varphi = \mathrm{id}_{M\otimes N}$. □

Lecture 3

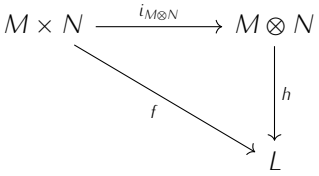
Proposition 2.4. Suppose M, N are R-modules, then

$$\sum_i m_i \otimes n_i = 0 \in M \otimes_R N$$

if and only if for all R-modules L, and every R-bilinear map $f : M \times N \to L$ has

$$\sum_i f(m_i, n_i) = 0$$

Proof. Suppose $\sum m_i \otimes n_i = 0$, let $f : M \times N \to L$ be bilinear. Then f factors through $M \times N \to M \otimes_R N$, and we can write



In this case, we have that

$$\sum_i f(m_i, n_i) = \sum_i h(i(m, n)) = \sum_i h(m_i \otimes n_i) = h\left(\sum_i m_i \otimes n_i\right) = h(0) = 0$$

Conversly, if

$$\sum_i m_i \otimes n_i \neq 0$$

then by definition,

$$\sum_i i_{m\otimes n}(m_i, n_i) \neq 0$$

□

Example 2.5

Let k be a field, and consider the tensor product

$$k^m \otimes k^\ell$$

Suppose k^m has basis $\{e_1, \ldots, e_m\}$, and k^ℓ has basis $\{f_1, \ldots, f_\ell\}$, then

$$k^m \otimes k^\ell = \operatorname{span}_k\{v \otimes w \mid v \in k^m, w \in k^\ell\} = \operatorname{span}_k\left\{e_i \otimes f_j\right\}$$

Claim 2.6. $\{e_i \otimes f_j\}$ is a basis.

Proof. Suppose we have

$$\sum_{ij} \alpha_{ij}(e_i \otimes f_j) = 0$$

For every $1 \leq a \leq m, 1 \leq b \leq \ell$, define a bilinear map

$$T_{ab} : k^m \times k^\ell \to k$$
$$T_{ab}((v_i), (w_j)) = v_a w_b$$

This is a k-bilinear map. By proposition 2.4,

$$0 = \sum_{i,j} \alpha_{ij} T_{ab}(e_i, f_j) = \sum_{i,j} \alpha_{ij} \delta_{ia} \delta_{jb} = \alpha_{ab}$$

□

Example 2.7

More concretely, let us consider $\mathbb{R}^2 \otimes \mathbb{R}^2$. We have a basis of size 4, given by

$$e_1 \otimes f_1, e_1 \otimes f_2, e_2 \otimes f_1, e_2 \otimes f_2$$

What do pure tensors look like?

$$(\alpha e_1 + \beta e_2) \otimes (\gamma f_1 + \delta f_2) = \alpha\gamma(e_1 \otimes f_1) + \alpha\delta(e_1 \otimes f_2) + \beta\gamma(e_2 \otimes f_1) + \beta\delta(e_2 \otimes f_2)$$

These are not generic elements of $\mathbb{R}^2 \otimes \mathbb{R}^2$, since the vectors

$$(\alpha\gamma, \alpha\delta) \quad \text{and} \quad (\beta\gamma, \beta\delta)$$

are linearly dependent. In particular,

$$e_1 \otimes f_1 + 2e_1 \otimes f_2 + 3e_2 \otimes f_1 + 4e_2 \otimes f_2$$

is *not* a pure tensor.

Example 2.8 (warning)

First consider

$$\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

In this case,

$$2 \otimes 1 = 1 \otimes 2 = 1 \otimes 0 = 0$$

Now consider

$$2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

But in this case,

$$2 \otimes 1 \neq 0$$

since we can define a bilinear map

$$B : 2\mathbb{Z} \times \mathbb{Z}/2 \to \mathbb{Z}/2$$
$$B(2m, x) = mx$$

In this case,

$$B(2, 1) = 1 \cdot 1 = 1 \neq 0$$

However, if $M' \leq M, N' \leq N$ are submodules, and

$$\sum_i m_i \otimes n_i = 0$$

in $M' \otimes N'$, then

$$\sum_i m_i \otimes n_i = 0$$

in $M \otimes N$.

Proposition 2.9. If

$$\sum m_i \otimes n_i = 0 \in M \otimes_R N$$

then there are finitely generated R-submodules $M' \leq M, N' \leq N$, such that

$$\sum m_i \otimes n_i = 0 \in M' \otimes_R N'$$

Intuitively, a proof that the sum is zero is finite, and so it can only involve finitely many expressions. We

can take them to be the generators.

Proof.

$$\sum m_i \otimes n_i 0 =\in M \otimes N = \frac{R^{\oplus(M\times N)}}{\mathcal{K}}$$

then

$$\sum_i e_{(m_i,n_i)} = 0 \in \mathcal{K}$$

This means that we can write the left hand side as a finite sum of the generators of $\mathcal{K}$. Taking all the elements of M and N which appear, gives the result. □

Corollary 2.10. Let A, B be torsion-free abelian groups, then $A \otimes_{\mathbb{Z}} B$ is torsion free.

Proof. Suppose

$$n \cdot \left(\sum_i a_i \otimes b_i \right) = 0 \in A \otimes B$$

for some $n \geq 1$. By proposition 2.9, there exists finitely generated subgroups $A' \leq A, B' \leq B$, such that

$$n \cdot \left(\sum_i a_i \otimes b_i \right) = 0 \in A' \otimes B'$$

By the structure theorem of finitely generated abelian groups, $A' \cong \mathbb{Z}^r, B' \cong \mathbb{Z}^s$, and so we have that

$$A' \otimes B' \cong \mathbb{Z}^{rs}$$

which is torsion free. Contradiction. □

Example 2.11

$$\mathbb{C}^2 \otimes_{\mathbb{C}} \mathbb{C}^3 \cong \mathbb{C}^6$$

as $\mathbb{C}$-vector spaces, and we also have that $\mathbb{C}^6 \cong \mathbb{R}^1 2$ as $\mathbb{R}$-vector spaces. On the other hand,

$$\mathbb{C}^2 \otimes_{\mathbb{R}} \mathbb{C}^3 \cong \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{R}^6 \cong \mathbb{R}^{24}$$

Proposition 2.12.

1. $M \otimes N \cong N \otimes N$
2. $(M \otimes N) \otimes P \cong M \otimes (N \otimes P) \cong M \otimes N \otimes P$, where we define $M \otimes N \otimes P$ using trilinear maps.
3. $\left(\bigoplus_i M_i\right) \otimes P \cong \bigoplus_i (M_i \otimes P)$
4. $R \otimes_R M \cong M$,

Proof. See examples sheet 1. □

Example 2.13

Using proposition 2.12, we can compute

$$\begin{aligned} R^m \otimes R^\ell &\cong \left(\oplus_{i=1}^m R\right) \otimes \left(\oplus_{j=1}^\ell R\right) \\ &\cong \oplus_{i,j} R \\ &\cong R^{m\ell} \end{aligned}$$

2.1 Tensor product of R-linear maps

Proposition 2.14. For R-linear maps $f : M \to M'$, $g : N \to N'$, then there exists a unique R-linear map

$$f \otimes g : M \otimes N \to M' \otimes N'$$

with

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

Proof. Uniqueness is clear since the pure tensors generate. For existence, we can use the universal property on the R-bilinear map

$$\begin{aligned} T : M \times N &\to M' \otimes N' \\ T(m, n) &= f(m) \otimes g(n) \end{aligned}$$

□

Index

algebra, 3
    finitely generated, 3
    homomorphism, 3
Artinian
    module, 1
    ring, 2

exact sequence, 2
    short exact sequence, 2

Hilbert basis theorem, 3

module, 1

Noetherian
    module, 1
    ring, 2

tensor product
    of R-linear maps, 9
    of R-modules, 4

universal property of
    tensor product, 5